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AI Engineer

Professional Summary

Data Scientist at Harbor Tech, developing two flagship products under Harbor AI 2.0: an Industrial Intelligent Monitoring Platform for ongoing equipment monitoring and diagnostics, and an Equipment End-of-Line Quality Inspection Platform for customer-specific unit-by-unit inspection. Led monitoring product/domain design and the inspection project from requirements and specifications through implementation, UI/UX, and on-site delivery. Built traceable issue/project records and parallel AI-assisted development, repair, and verification workflows, supported by daily functional regression and human release review.

My research background includes multi-scale CNNs for DNA-sequence-based chromatin interaction prediction, passenger-flow prediction, person re-identification, spatiotemporal analysis, and recommendation systems.

Technical Skills

- **Industrial AI and Validation:** predictive maintenance, RUL, PRPD/vibration, FFT/WAV, anomaly detection, ISO 20816, ablations, confidence/coverage, and data-availability boundaries.
- **AI/LLM Systems:** LangChain/LangGraph, report-service integration, vLLM, OpenAI-compatible APIs, deterministic facts, fallbacks, per-device isolation, and human validation.
- **Data Science and Research:** Python, pandas, NumPy, SciPy, scikit-learn, CNNs, multi-scale models, cross-dataset evaluation, explainable AI (XAI), statistical tests, and effect sizes.
- **Backend and Platform:** Django/Django Ninja, Pydantic, PostgreSQL, Redis, RabbitMQ, REST APIs, asynchronous workflows, repository patterns, and artifact-by-reference.
- **Product and Delivery:** React/TypeScript, Docker/Docker Compose, offline and cross-architecture installation, release/rollback, migration, Git, automated tests, AI agents, and human review.

Work Experience

Data Scientist, Harbor Tech | Taipei, Taiwan | Oct 2025-Present

Harbor AI 2.0 comprises two flagship products with distinct purposes and shared underlying technology: ongoing equipment monitoring and diagnostics, and customer-specific end-of-line quality inspection. Led product/domain design, API and data contracts, analytic-service integration, and delivery integration for monitoring; led the inspection project from customer requirements to on-site delivery. The workstreams below describe requirements, my decisions, established capabilities, and completion boundaries; AI agents and engineering partners contributed according to the specific function.

Harbor AI 2.0 Equipment End-of-Line Quality Inspection Platform | End-to-End Project, Ongoing

- Led my first end-to-end project from customer requirements and specifications through system implementation, UI/UX design and refinement, and on-site delivery. Used shared underlying technology to establish a distinct product for fixtures inspecting successive manufactured units, defining units under test, measurement batches, data relationships, decisions, and reports.

- Clarified synchronized and switched measurement points, good-unit baseline rerecording, rush orders, serial numbers, and work-order continuity, connecting inspection steps to data selection, decisions, human review, and reports. Kept functional tests separate from thresholds needing field calibration.
- The initial AI-generated UI met the specification and functional acceptance criteria, but operation required roughly five to six page levels, repeated returns, and circular navigation. Reorganized the work into three core interface types: plan list, work order, and board. Consolidated reports, rechecks, and baseline management or moved them into dialogs. DQA feedback was qualitatively positive; operation time and error rates were not measured.
- Placed previous/next/primary actions near the bottom center and supported arrow keys and Enter for repeated operation. Moved infrequent or accidental-trigger-sensitive actions to the bottom left, and refined type sizes, status visibility, and sensor-detachment alerts.
- Separated sensor-installation confirmation from measurement, using sensor IDs and installation diagrams as mandatory checks before measurement to prevent incorrect sensor installation.
- Coordinated the Phase 1 quality-testing (DQA) handoff, test versions, manuals exercised on the test machine, and acceptance records. In September, query, unit-switching, human-review, and export work entered a local integration branch.
- On October 12, 2026, I attended Phase 1 on-site acceptance activities, demonstrated the software, provided user training, and brought back customer feedback. Formal sign-off, the final acceptance conclusion, and the latest mainline merge/release status have not been confirmed. I led the work across requirements, specifications, system implementation, UI/UX refinement, and delivery, using AI agents and collaborating with engineering partners on implementation and verification.
- I estimate my overall project contribution at about 80%, across the end-to-end work described above. This is a personal estimate of total contribution, not a measured share of code written.

Harbor AI 2.0 Industrial Intelligent Monitoring Platform | Product and Domain Architecture

- Led 2.0 monitoring product/domain design in an environment with an existing 1.0 system. Built most 2.0 capabilities from scratch with the team while evolving the architecture and UI/UX. Organized reporting, RUL, multi-sensor, trend, partial-discharge, and threshold-update requirements into functional categories, user flows, and validation criteria for ongoing monitoring, investigation, and diagnostics.
- Separated vibration and partial-discharge meanings for RUL, trends, PRPD status, and data quality into layered sensor status. Used a device-first report flow to reduce incorrect device or data selection.
- Contributed the main frontend implementation, connecting Report, Sensor Status, PD Event Analysis, RUL, and scheduling flows to real APIs. Supported multilingual interfaces, RUL charts, mobile layouts, and sensor-detail workflows, with shared pages attributed according to team responsibilities.

From Coexistence and Gradual Replacement to Independent 2.0 Operation

- **Field ownership and compatibility:** In March 2026, addressed periodic synchronization between the legacy and new systems. Distinguished legacy-provided equipment data from 2.0 user settings, protecting new-system settings from silent overwrites and turning coexistence conflicts into explicit data contracts.
- **Device authority and gradual replacement:** In July, advanced the transfer of equipment control, authorization, and management to 2.0, narrowing the legacy role to device discovery and unifying state-transition rules. In August, decided to move event analysis to native 2.0 events and retired that function's legacy source after validating the analysis-to-UI chain. That result applied to the specific function at the time.
- **Auditable data migration:** In July, consolidated legacy ingestion into tools and data contracts with manifests, audits, and explicit failures. I required explicit sensor-number mapping: missing or ambiguous mappings stop the process, requiring the manifest and mapping to be corrected before revalidation rather

than guessing. I advanced the corresponding product and migration integration; AI agents and engineering partners contributed implementation and verification.

- **Current state, personally confirmed on October 23, 2026:** 2.0 no longer depends on 1.0 and can operate independently. The dates above document individual stages, not the date of full dependency removal. This confirmation does not establish that all customers switched, legacy services shut down, or all historical data migrated.

Human-AI Development, Daily Regression, and Project Records

- Built a shared issue/project-record framework linking problems, analysis, decisions, implementation results, verification, and versions. This made changing requirements and parallel work traceable for subsequent agents and human reviewers.
- Used the same workflow for new features and repairs. I decomposed requirements, organized work, supplied persistent context and tools, and defined responsibilities, validation criteria, and stop conditions. AI agents researched, implemented, and tested on isolated branches, allowing work and verification to proceed in parallel.
- Ran daily automated functional regression and monitoring on the development machine, retaining test plans, results, and anomalies. Findings created or updated issues; AI agents repaired and rechecked changes on isolated branches before submitting Bitbucket PRs for human review.
- Reviewed domain assumptions, data provenance, and product behavior. Reviewers and I used change and validation evidence to decide merge and release; a PR or passing tests supplied review evidence rather than release approval.
- Fed recurring problems back into reusable work rules, tests, and checks. Used VitePress to connect code, issues, versions, manuals, and management summaries. Established a workflow that can be continued and inspected; automatic-repair success rates, test-coverage percentages, and productivity multipliers have not been measured.

Overview Monitoring and Event Semantics

- Defined overview status rules that distinguish unresolved-event severity, device connectivity, and data eligible for health calculations, allowing users to trace equipment and events needing attention.
- Designed overview updates and notification behavior, integrating state notifications and throttling into existing SSE infrastructure. I defined product behavior, evaluated tradeoffs, and set validation criteria; AI agents supported implementation and verification.
- Contributed event-window, waveform-availability, diagnostic-dispatch, and result-linking design. Kept platform-side functions distinct from Gateway recording integration and incomplete field end-to-end validation.

RUL, Diagnostics, and Model-Assumption Validation

My responsibility was to review model assumptions, data provenance, and product presentation together, establishing reproducible analysis and rules for insufficient evidence. The test-data example below shows where an AI conclusion required my judgment and a fresh validation plan.

- Audited AI-generated RUL hypotheses and data provenance. Traced the data source and found test constants misrepresented as observed data. Invalidated the conclusion, then required multi-parameter, multi-curve, and forward/reverse ablations to recheck the assumptions.
- Connected research notebooks directly to production implementations, adding source comparisons to reduce research-to-code drift. Set quality criteria for both missed rapid degradation and false alarms on healthy sequences.

- Established a responsibility chain from preprocessing through vibration-activity screening (VAD), anomaly classification, ISO severity, RUL, and reporting, separating fault type, severity, and remaining time.
- Reviewed segment sampling, fairness, coverage, selection bias, and denominators. Marked results as degraded, interval-only, or unavailable when evidence was insufficient, retained raw engine outputs, and prevented UI or LLM layers from overwriting model facts. Field lifetime ground truth and confidence calibration remain incomplete; no RUL-accuracy claim is made.

Platform APIs, Asynchronous Workflows, and Artifact Contracts

- Collaborated with platform engineers on Django/Django Ninja workflow orchestration, defining product behavior, APIs, repository interfaces, storage routing, and artifact contracts across diagnostic, RUL, and report services.
- Used backpressure, idempotency, timeouts, asynchronous callbacks, and task lifecycles to check interfaces and failure behavior. My scope covered service contracts and validation; the underlying runtime and its failure-handling implementation were shared team responsibilities.
- Used artifact-by-reference for cross-service outputs, defining access and backup boundaries and reducing direct transfer of large payloads between tasks.
- Integrated database-authoritative per-sensor scheduling, manual triggers, batches, and execution history. Connected sensor updates to RUL service results and history, routing workflows through aggregate data or WAV inputs according to availability.

Edge Ingestion, Software OTA, and Threshold Control

- Owned Gateway ingestion contracts for bundled data upload, heartbeats, asynchronous processing, deduplication, integrity, capacity, and security. Led system design, option validation, cross-team alignment, and handoff; the responsible engineer owned subsequent server implementation.
- Contributed Gateway software-package registration, storage, version lookup, download, architecture compatibility, and integrity verification. The work focused on server-side package distribution.
- Designed and integrated feedback from RUL recommendations through platform threshold state/history to Gateway threshold updates, including per-sensor automatic/manual policy, emergency disablement, scheduling, and fault isolation.
- Kept online ingestion, periodic legacy-system synchronization, offline migration, Gateway software updates, and RUL threshold feedback distinct, each with its own contracts and validation conditions.

Docker, Offline Deployment, Updates, and Migration

- Integrated Docker/offline image packages, cross-architecture installation, installation state, and release metadata into repeatable, traceable delivery workflows.
- Built platform offline-update and recovery paths, isolating update execution from the platform's own shutdown risk and adding preflight checks, progress/history, failure signals, and verifiable rollback.
- Migration validation also checked fields, time zones, readability, continuity, and legacy/new-system coexistence conditions, with recovery procedures retained.
- Paused writes and added fresh schemas, constraint checks, and rollback procedures after real-machine restore conflicts. Reviewed development/production serving, storage, restart, and deployment constraints, incorporating gaps into initialization, upgrades, and automated checks.
- Reduced delivery-package scope, removed accidentally included customer raw data, completed fresh-install validation, and led integration and cut decisions for two shipped releases. Feature implementation remains attributed to engineering partners and AI agents according to their contributions.

From Customer Report Skills to Platform Report Services

- Started with customer analysis and manually reviewed report delivery, extracting reusable diagnostic steps, report skills, quality criteria, and human-review rules. Used research notebooks to check analysis and report correctness.
- In July 2026, used a LangChain/LangGraph POC to validate state, node contracts, per-device isolation, partial regeneration, fallbacks, and artifacts. Deterministic nodes produced values and classifications; the LLM organized narrative without changing facts.
- In August, moved the report-generation path into a service connected to the platform. In September, DOCX reports and frontend download integration joined the mainline. I led methods, domain interpretation, and quality conditions; AI agents and engineers contributed implementation, testing, and integration.
- Delivered in stages: July POC validation, subsequent report-path service integration, and DOCX/download integration. Field orchestration, deployment, and customer acceptance require separate verification.
- Validated a vLLM OpenAI-compatible provider against a live endpoint for chat, embeddings, tool calling, and connectivity/capability/semantic health checks. Validation covered provider endpoint capabilities; rollout status was tracked separately for each report path.

Field Signal Analysis and Customer Delivery

- Across semiconductor-manufacturing and large multi-sensor field-analysis contexts, aligned field and quality teams on partial-discharge configurations, DB/WAV formats, event scope, and handoff rules. Combined full database context with event-window WAV selection rather than transferring and analyzing every raw file.
- Connected environment checks, event selection, WAV extraction, batch analysis, accumulated PRPD images, dashboards, AI drafts, and human validation. Required page-by-page human review before low-resolution per-second charts could support customer conclusions.
- Reverse-engineered legacy code, notebooks, and reports to rebuild traceable ADC-to-g, FFT, SNR, cutting-point, and frequency-domain analysis. Delivered executables and user guides while protecting company algorithm IP.
- In a motor/pump vibration POC, used max/P99 to distinguish sustained severity from one-second transient spikes. Checked operating segments, RUL basis, fit constraints, and VRMS volatility, reporting ISO severity separately from low-confidence RUL.
- In a later current/vibration analysis, used minute medians to reduce autocorrelation, nonparametric tests, multiple-comparison correction, and effect sizes to narrow stakeholder observations to partial or sensor-specific support. Delivered the analysis and customer explanation. Root cause still requires field information such as VFD/RPM, flow, head, valve position, and BEP; statistical differences alone do not establish cause.

Technical Mentoring | Summer 2026

- Defined offline research topics, division-of-work principles, vibration and phase-resolved partial-discharge primers, ISO 13374/20816 material, starter notebooks, and algorithm specifications for a research intern.
- Reviewed progress and research outputs through weekly reports and realigned tasks when data assumptions drifted. Provided mentoring and research coordination; the intern owned their analysis results.

Smart-City Edge-Platform Technical Proposal | Unsubmitted Draft

- Owned company-side technical drafting, version convergence, and cross-company interface/checkpoint alignment, standardizing platform terminology, data/control planes, and responsibilities.
- Refined schedules and checkpoints, resolved document inconsistencies, and set review criteria for source reliability, claim boundaries, freedom-to-operate (FTO), and KPI risks. This remains an unsubmitted planning draft, without approval, award, implementation, or deployment.

Data Science Intern, Institute of Sociology, Academia Sinica | Taipei, Taiwan | May 2019-Dec 2020

- Built a social-survey (TSCS) survey-area dispersion and route-planning workflow combining sampled addresses, point-to-point and fallback distances, survey-area grouping, OR-Tools TSP, and Folium for field surveys, routes, and allowance/budget estimation.
- Used Selenium, multiprocessing, Tor multi-port routing, and Folium for distance collection and GIS route output; these describe the project's technology at that time.

Representative Research and Projects

MCIENet | M.S. Thesis, 2024

- Predicted 3D chromatin interactions/CTCF loops from DNA sequences using multi-scale CNN/Inception architecture, examining signals from different receptive fields.
- Independently led research decomposition, model and data-pipeline integration, experiment design/execution, ablations, cross-dataset evaluation, XAI, thesis writing, and reproducible repository engineering.
- Used BaseCNN comparisons, sequence-length settings, and ablation configurations to examine design choices. Retained DeepLIFT, ChINN/DNABERT2 comparison entry points, and shared split configurations for traceability. Findings are limited to completed experiments; comparison entry points and shared configurations were retained for further reproduction.
- Treated dataset, positional, and resolution differences as generalization checks, retaining cross-cell-line limitations. Adapted parts of positive/negative preprocessing from ChINN and distinguished that prior work from my model, experimental, and engineering contributions.

Financial-Text Sentiment Analysis | TCSE Research, 2020

- Studied stock-article sentiment using a Hierarchical Attention Network and sentiment lexicons. The work began with an approved undergraduate research grant and led to a TCSE publication.
- Performed all research work for the study; the paper was coauthored, so this scope does not imply sole authorship.
- Early repeated model changes yielded limited improvement. Gradually added data and parameter comparisons, retaining comparison conditions and experimental results as the basis for decisions.

Dynamic Embedding for Passenger-Flow Estimation | IIAI AAI Research, 2021

- Explored dynamic embedding for passenger-flow estimation. Responsible for method ideation, data processing, model construction, and experiment execution.
- Used data and parameter comparisons to examine model choices. The research received an IIAI AAI Competitive Paper Award; publication credit remains shared.

Research Experiment Practices Developed Over Time | 2020-2024

- Progressed from repeated model changes with limited improvement toward data and parameter comparisons, then more complete ablation design and parameter-adjustment logic in the MCIENet thesis.
- Used Shell to organize parallel experiments under limited GPU resources, including running two parameter configurations on one GPU with separate logs and checkpoints.
- Saved preprocessed data as H5, used early stopping, and first sampled three parameter values to inspect the curve before deciding how to adjust the search.
- Developed these practices across the 2020 sentiment study, 2021 passenger-flow study, PropGMAN, and MCIENet, with different levels of use in each project. This does not imply an identical complete pipeline across all projects or a measured overall speedup.

Dynamic Traffic Network Representation | KBS Collaborative Research, 2024

- Coauthored research on dynamic traffic network representation for passenger-flow prediction. My actual contribution was primarily supporting code and technical assistance.

Horus Human Identification System | Undergraduate Capstone, 2020

- The team integrated multi-object tracking, person re-identification, and face recognition into a multi-camera identification and tracking system.
- Primarily owned ReID feature-extraction integration, folder sampling/matching, cosine similarity, dual-threshold decisions, and tracking-record queue orchestration.
- Integrated the third-party FastReID base model with the team' s MOT and face-recognition modules. The team won first place in Soochow University's undergraduate capstone competition.

PropGMAN Asset Activation | Team Project, 2022

- In a three-person team, applied spatiotemporal modeling to property-transaction records to predict regional land prices and transaction volumes for asset-activation analysis.
- Proposed the project direction, led team coordination and division of work, and handled model implementation, experiments, and the final-round report. Retained preprocessing, model settings, experimental results, and reproduction instructions. The base GMAN method used an existing third-party research implementation. The team received a Creative Award in the AI for Asset Activation Innovation Contest.

Additional Independent Projects

Dynamic Bus Route and Schedule Planning | Project period to be confirmed

- Integrated tourism demand, Google Trends, Popular Times, attraction locations, and travel distances for route and schedule planning in eastern Taiwan. Built demand-allocation and distance-matrix workflows, explored OR-Tools TSP routes, and visualized routes and schedules with Folium, Gantt, and Sankey views.
- This was an exploratory independent project, separate from the Academia Sinica survey-routing work. The project period remains unconfirmed; no transit-operator adoption or operational outcome is claimed.

Additional Team Projects

- **Distributed Spark Crawler/Recovery, 2022:** An NCCU distributed-systems course team connected collection, missing-data detection, Spark-based imputation, a database, and dashboards to demonstrate recovery from collection failures. I owned the Spark cluster, GitHub work, and demo video, and contributed system integration.

- **Financial Recommendation and Automated Notifications (E.SUN Fugle), 2020:** Proposed the topic direction and led team coordination and division of work. Primarily designed, implemented, and deployed the Telegram chatbot, integrating modules from teammates into one entry point. The team used item-based, user-based, and co-occurrence recommendation approaches; I offered algorithm ideas and suggestions rather than implementing every algorithm myself. The team received an Excellence Award in a financial-programming course competition.
- **Taipei Disaster Resource Dashboard, 2025:** Proposed the topic and collected data for a team analysis of disaster-prevention resources around tourist attractions. The team reached the Taipei-New Taipei City Dashboard Hackathon finals. The public dashboard is based on a fork of a friend' s template, with team and template credit retained.

Research and Publications

- Dynamic traffic network representation model for improving the prediction performance of passenger flow. Knowledge-Based Systems, 2024. Primarily contributed supporting code and technical assistance.
- A Dynamic Embedding Method for Passenger Flow Estimation. IIAI AAI, 2021. Responsible for method ideation, data processing, model construction, and experiments. Competitive Paper Award.
- Hierarchical Attention Network with Sentiment Lexicons for Stock Article Emotion Analysis. TCSE, 2020. Performed all research work; the paper was coauthored. Preceded by an approved undergraduate research grant.
- MCIENet: Multi-scale CNN-based Information Extraction from DNA Sequences for 3D Chromatin Interactions Prediction. M.S. thesis, 2024.

Education

- National Chengchi University, M.S. in Computer Science | Sep 2021-Sep 2024.
- Soochow University, B.S. in Data Science | Sep 2017-Jul 2021.

Recognition

- IIAI AAI Competitive Paper Award, 2021.
- Creative Award, AI for Asset Activation Innovation Contest, 2022.
- First Place, Soochow University Undergraduate Capstone, Horus, 2020.
- Excellence Award, Financial Programming Course Competition, 2020.
- Finalist, Taipei-New Taipei City Dashboard Hackathon, 2025.